

Name \_\_\_\_\_

Period \_\_\_\_\_ Date \_\_\_\_\_

**Practice Test – Advanced Kinematics (Version 3)**

*On any problems, you must show all work, include units, round properly, and box your final answer.*

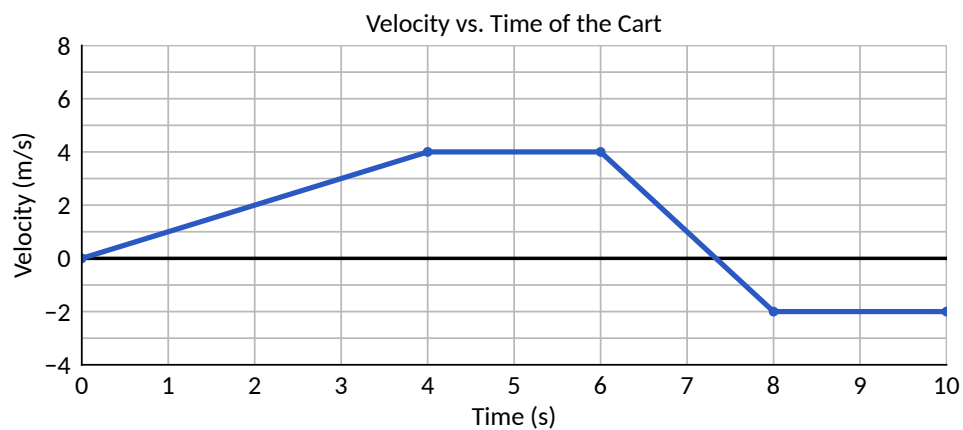
*Do the algebra first, before plugging in values.*

**Rewrite the following statement in the space below:** On my honor, before God, I have neither given, nor received, any aid on this test.

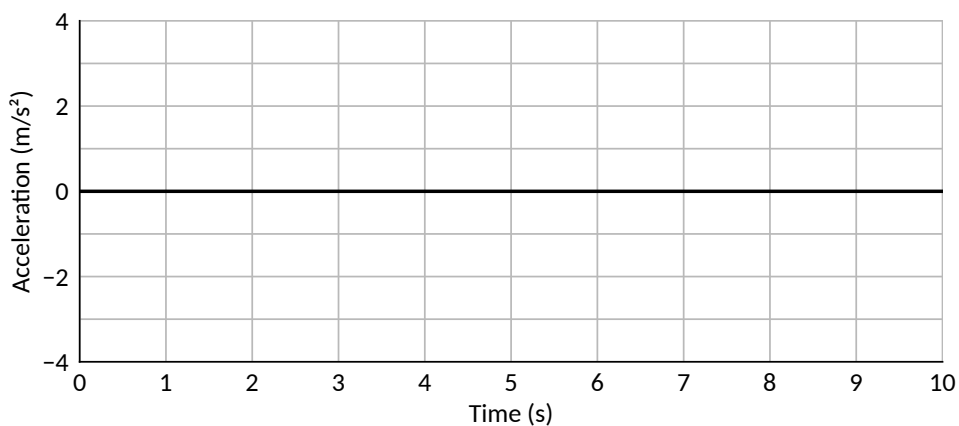
**For all questions:**

- Use  $g = 9.80 \text{ m/s}^2$  and neglect air resistance.
- Round numerical answers to three significant figures.
- 50 minutes. Four questions, 25 points each (100 points total).
- The last page is optional extra practice and is not part of the timed test.

**1. (25 points)** The velocity-time graph below shows the motion of a cart moving along a straight track from  $t = 0$  s to  $t = 10$  s. On the blank grid, draw the corresponding acceleration-time graph. Show the slope calculation for each segment of the velocity-time graph.



Draw your acceleration-time graph here:



Slope calculations:

**2. (25 points)** A car travels along a straight, level road at a constant 30.0 m/s. The driver then brakes, and the car slows with a constant acceleration of magnitude  $7.50 \text{ m/s}^2$  until it stops. How far does the car travel while braking?

**3. (25 points)** A football is kicked from ground level on a level field with an initial speed of 28.0 m/s at  $30.0^\circ$  above the horizontal. It lands at the same height from which it was kicked. Air resistance is negligible. What is the ball's range (the horizontal distance it travels before landing)?

**4. (25 points)** A small object of mass  $m$  is released from rest at the top of a skyscraper and falls straight down a vertical distance  $d$  to the ground, as shown in Figure 1. Air resistance is negligible. Take downward as the positive direction.

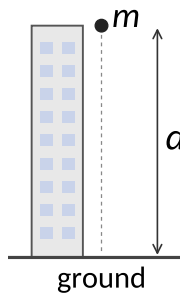


Figure 1

**A.** Derive an expression for  $v_y$ , the velocity of the object just before it reaches the ground. Express your answer in terms of  $m$ ,  $d$ , and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

**B.** The same object is released from rest from the same height on a planet where the acceleration due to gravity is one-fourth of its value on Earth. A student claims that the object would reach the ground with half the speed it has on Earth. Indicate whether your expression from part A is or is not consistent with this claim. Briefly justify your answer by referencing your derivation in part A.

**Optional extra practice (not part of the timed test, not scored)**

**E1.** For the football in Question 3: (a) What is its hang time? (b) What is its maximum height?

**E2.** For the car in Question 2: (a) How long does it take to stop? (b) If its initial speed were doubled and it braked with the same acceleration, by what factor would its braking distance change?

**E3.** For the object in Question 4: write its velocity  $v_y(t)$  as a function of the time  $t$  since it was released, and state the time interval over which this expression is valid.

## Answer Key – Practice Test, Version 3 – Advanced Kinematics

Worked step by step, the same way as the class notes: sketch, list the variables, pick the equation, algebra first, then numbers with units.  $g = 9.80 \text{ m/s}^2$ ; answers to three significant figures.

**AP Physics 1 equation sheet, kinematics (Mechanics and Fluids, first three):**

**#1**  $v_x = v_{x0} + a_x t$  · **#2**  $x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$  · **#3**  $v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$

Same equations as  $v_f = v_i + at$ ,  $\Delta x = v_i t + \frac{1}{2} at^2$ ,  $v_f^2 = v_i^2 + 2a\Delta x$ . Pick the one that leaves out the variable you don't know and don't need. For vertical motion use  $y$  in place of  $x$ .

### 1. Acceleration-time graph (25 points)

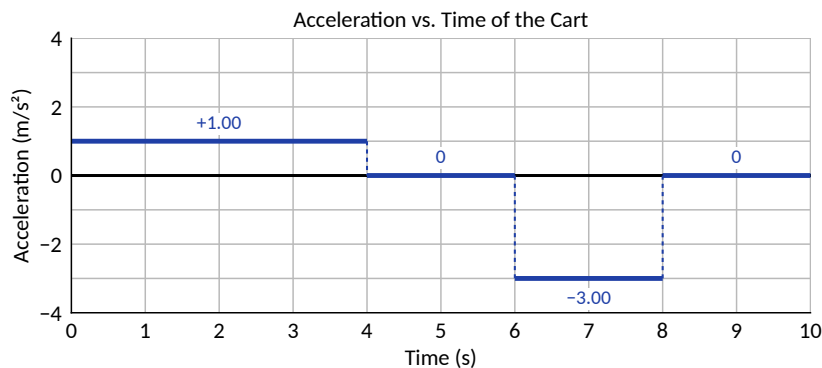
**Step 1** Acceleration is the slope of the v-t graph.

$$a = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i} \text{ (read } v_i \text{ and } v_f \text{ at the start and end of each straight piece; values read off grid lines are treated as exact)}$$

**Step 2** Find the slope of each segment.

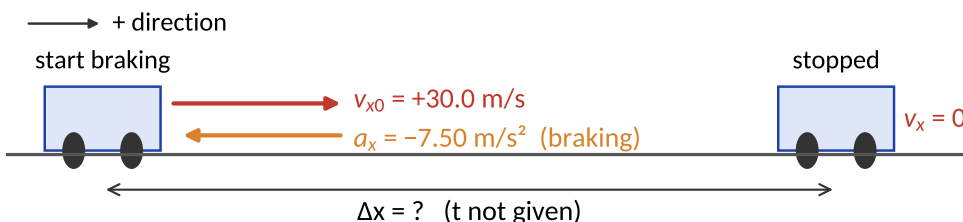
| segment     | $\Delta v = v_f - v_i$      | $\Delta t = t_f - t_i$ | $a = \Delta v / \Delta t$ |
|-------------|-----------------------------|------------------------|---------------------------|
| 0 s to 4 s  | 4 m/s - 0 m/s = 4 m/s       | 4 s - 0 s = 4 s        | +1.00 m/s <sup>2</sup>    |
| 4 s to 6 s  | 4 m/s - 4 m/s = 0 m/s       | 6 s - 4 s = 2 s        | 0 m/s <sup>2</sup>        |
| 6 s to 8 s  | (-2 m/s) - 4 m/s = -6 m/s   | 8 s - 6 s = 2 s        | -3.00 m/s <sup>2</sup>    |
| 8 s to 10 s | (-2 m/s) - (-2 m/s) = 0 m/s | 10 s - 8 s = 2 s       | 0 m/s <sup>2</sup>        |

**Step 3** Draw each slope as a flat line over its time interval. Label both axes with units.



Watch out: the cart turns around where  $v$  crosses 0, but the slope does not change there, so  $a$  does not change either. A straight slanted v-t line always becomes a flat a-t line; a flat v-t line becomes  $a = 0$ .

## 2. Braking car: stopping distance (25 points)



### Step 1 Pick a positive direction and get the sign of a.

Let the direction the car moves be +. The car is moving + but slowing down, so its velocity is decreasing:  $\Delta v$  is negative, which makes a negative. Braking acceleration points opposite the motion, so  $a = -7.50 \text{ m/s}^2$ .

### Step 2 List the five variables.

| $v_i (= v_{x0})$ | $v_f (= v_x)$ | $a$                    | $t$                  | $\Delta x$ |
|------------------|---------------|------------------------|----------------------|------------|
| +30.0 m/s        | 0 m/s (stops) | -7.50 m/s <sup>2</sup> | not given, not asked | ? (wanted) |

### Step 3 Pick the equation.

| AP sheet | equation                                   | leaves out                |
|----------|--------------------------------------------|---------------------------|
| #1       | $v_x = v_{x0} + a_x t$                     | $x - x_0$ ( $\Delta x$ )  |
| #2       | $x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$ | $v_x$ (final velocity)    |
| #3       | $v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$         | $t$ (time) ← use this one |

**Why #3:** time is not given and not asked for. #3, written  $v_f^2 = v_i^2 + 2a\Delta x$ , is the only one without  $t$ .

### Step 4 Start from $v_f^2 = v_i^2 + 2a\Delta x$ and set $v_f = 0$ . Solve for $\Delta x$ before any numbers.

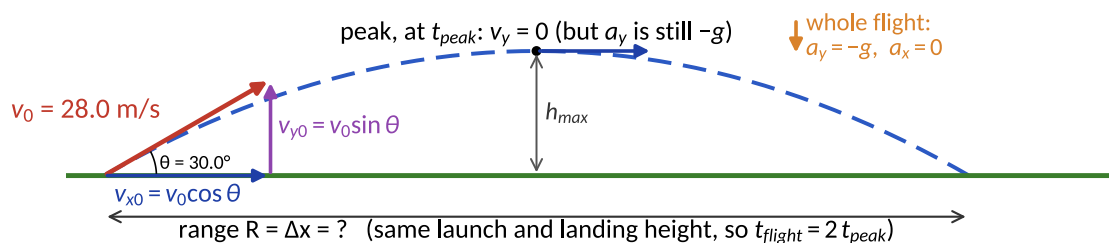
$$v_f^2 = v_i^2 + 2a\Delta x \rightarrow 0 = v_i^2 + 2a\Delta x \rightarrow 2a\Delta x = -v_i^2 \rightarrow \Delta x = -\frac{v_i^2}{2a}$$

### Step 5 Plug in numbers with units.

$$\Delta x = -\frac{(+30.0 \text{ m/s})^2}{2(-7.50 \text{ m/s}^2)} = -\frac{(900 \text{ m}^2/\text{s}^2)}{(-15.0 \text{ m/s}^2)} = \boxed{60.0 \text{ m}}$$

The minus sign in the formula and the negative  $a$  cancel, so  $\Delta x$  comes out positive: the car moves forward while it stops. If you get a negative distance,  $a$  had the wrong sign. Units:  $(\text{m}^2/\text{s}^2) \div (\text{m}/\text{s}^2) = \text{m}$ .

### 3. Football: range (25 points)



Split the kick into two motions. Sideways moves at a constant speed ( $a_x = 0$ ). Up and down is free fall ( $a_y = -g$ ). **Time** is the only thing both share.

**Step 1** Resolve the initial velocity into components (keep letters; numbers go in at the end).

$v_{x0} = v_0 \cos \theta$  (the side next to the angle: along the ground)

$v_{y0} = v_0 \sin \theta$  (the side across from the angle: straight up)

**Step 2** Make the T-chart.

| horizontal                 | vertical                         |
|----------------------------|----------------------------------|
| $a_x = 0$                  | $a_y = -g = -9.80 \text{ m/s}^2$ |
| $v_{x0} = v_0 \cos \theta$ | $v_{y0} = v_0 \sin \theta$       |
| $\Delta x = R = ?$         | at the peak: $v_y = 0$           |
| $t = t_{\text{flight}}$    | $t_{\text{peak}} = ?$            |

**Step 3** Find the time to the peak, where  $v_y = 0$ .

| AP sheet | equation                                   | leaves out                                          |
|----------|--------------------------------------------|-----------------------------------------------------|
| #1       | $v_y = v_{y0} + a_y t$                     | $y - y_0 (\Delta y) \leftarrow \text{use this one}$ |
| #2       | $y = y_0 + v_{y0} t + \frac{1}{2} a_y t^2$ | $v_y$ (final velocity)                              |
| #3       | $v_y^2 = v_{y0}^2 + 2 a_y (y - y_0)$       | $t$ (time)                                          |

**Why #1:** at the peak you know  $v_y = 0$ ,  $v_{y0}$  and  $a_y$ , and you want  $t$ . The height of the peak is not known, so use #1, the one that leaves out  $\Delta y$ .

$$v_y = v_{y0} + a_y t \rightarrow 0 = v_0 \sin \theta - g t_{\text{peak}} \rightarrow t_{\text{peak}} = \frac{v_0 \sin \theta}{g}$$

**Step 4 Double it for the whole flight.**

The ball lands at the same height it was kicked from, so the trip down takes exactly as long as the trip

$$\text{up: } t_{\text{flight}} = 2t_{\text{peak}} = \frac{2v_0 \sin \theta}{g}$$

Watch out: at the peak  $v_y = 0$ , but the acceleration is still  $a_y = -g$ , pointing down. Gravity never switches off. Doubling  $t_{\text{peak}}$  only works because the launch and landing heights match. And  $\Delta y = 0$  over the whole flight means it ends at the height it started, not that the acceleration is zero.

**Step 5 Use the flight time in the horizontal column.**

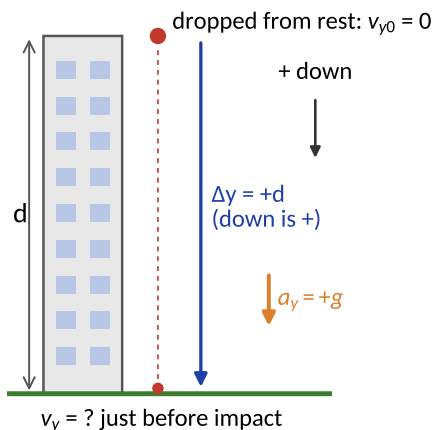
**Why #2 (x-form):** horizontally you know  $t$  and  $v_{x0}$  and want  $\Delta x$ ; with  $a_x = 0$ , #2 becomes  $\Delta x = v_{x0}t$ .

$$x - x_0 = v_{x0}t + \frac{1}{2}a_x t^2 \rightarrow R = v_{x0}t_{\text{flight}} = (v_0 \cos \theta) \frac{2v_0 \sin \theta}{g} \rightarrow \boxed{R = \frac{2v_0^2 \sin \theta \cos \theta}{g}}$$

**Step 6 Now plug in numbers with units.**

$$R = \frac{2(28.0 \text{ m/s})^2(\sin 30.0^\circ)(\cos 30.0^\circ)}{(9.80 \text{ m/s}^2)} = \boxed{69.3 \text{ m}}$$

Check:  $t_{\text{peak}} = (28.0 \text{ m/s})(\sin 30.0^\circ)/(9.80 \text{ m/s}^2) = 1.429 \text{ s}$ ,  $t_{\text{flight}} = 2.857 \text{ s}$ ,  $v_{x0} = 24.25 \text{ m/s}$ , and  $(24.25 \text{ m/s})(2.857 \text{ s}) = 69.28 \text{ m} \approx 69.3 \text{ m}$ . ✓ (2 sin  $\theta$  cos  $\theta$  = sin  $2\theta$ , so this also equals  $v_0^2 \sin 2\theta / g$ .)

**4. Dropped from a skyscraper, AP style (25 points): A. derive (18), B. justify with it (7)**

**Reading the AP wording.** “Physical constants” means  $g$ : you’re expected to use it even though it’s not listed. “As appropriate” means you may not need every letter listed. Here  $m$  drops out, because every object in free fall has the same acceleration  $g$ , whatever its mass.

Down is + (given), so the signs flip from the usual setup. The object ends  $d$  below the start, the + direction:  $\Delta y = +d$ . Gravity also points down:  $a_y = +g$ .

**A. Derivation (18 points)****Step 1 List the variables (vertical, so  $y$  in place of  $x$ ).**

| $v_{y0}$ (initial)     | $v_y$ (final) | $a_y$ | $t$                  | $y - y_0$ |
|------------------------|---------------|-------|----------------------|-----------|
| 0 (released from rest) | ? (wanted)    | +g    | not given, not asked | +d        |

**Step 2** Write the equation from the reference information first.

| AP sheet | equation                                   | leaves out                     |
|----------|--------------------------------------------|--------------------------------|
| #1       | $v_y = v_{y0} + a_y t$                     | $y - y_0$ ( $\Delta y$ )       |
| #2       | $y = y_0 + v_{y0} t + \frac{1}{2} a_y t^2$ | $v_y$ (final velocity)         |
| #3       | $v_y^2 = v_{y0}^2 + 2a_y(y - y_0)$         | <b>t (time) ← use this one</b> |

**Why #3:** t is not given and not asked. On the AP exam, write this equation on its own line before substituting anything; that line is where the first point is.

Start:  $v_y^2 = v_{y0}^2 + 2a_y(y - y_0)$

**Step 3** Substitute and solve with letters.

$$v_y^2 = 0^2 + 2(+g)(+d) = 2gd \rightarrow v_y = \pm\sqrt{2gd}$$

**Step 4** Pick the root.

It is moving **down** at the ground, and down is + here, so take the positive root:  $v_y = +\sqrt{2gd}$  No m in it. Units:  $\sqrt{(m/s^2)(m)} = m/s$ .

Why not  $-\sqrt{2gd}$ ? With down as +, that would mean moving up. Don't add a minus just because things fall; the sign comes from the convention.

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## B. Justify with your derivation (7 points)

**Consistent.** From part A, speed =  $\sqrt{2gd}$ , so speed grows with the **square root** of g. With g/4:

$$\sqrt{2(g/4)d} = \sqrt{1/4 \cdot 2gd} = \frac{1}{2}\sqrt{2gd}. \text{ One-fourth the gravity gives half the speed.}$$

## Optional extra practice (not scored)

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### E1. Football: hang time and maximum height

$$(a) \text{ Hang time} = t_{\text{flight}} = \frac{2v_0 \sin \theta}{g} = \frac{2(28.0 \text{ m/s})(\sin 30.0^\circ)}{(9.80 \text{ m/s}^2)} = \boxed{2.86 \text{ s}}$$

**Why #3 (y-form) for (b):** at the peak  $v_y = 0$ , and you want the height without needing  $t$ .

$$(b) 0 = (v_0 \sin \theta)^2 - 2g h_{\text{max}} \rightarrow h_{\text{max}} = \frac{v_0^2 \sin^2 \theta}{2g} = \frac{(28.0 \text{ m/s})^2 (\sin 30.0^\circ)^2}{2(9.80 \text{ m/s}^2)} = \boxed{10.0 \text{ m}}$$

### E2. Car: stopping time and the doubling question

**Why #1:** now  $t$  is wanted and  $\Delta x$  is not needed.  $v_f = v_i + at \rightarrow 0 = v_i + at \rightarrow t = -\frac{v_i}{a} = -\frac{(+30.0 \text{ m/s})}{(-7.50 \text{ m/s}^2)} =$

$$\boxed{4.00 \text{ s}}$$

(b)  $\Delta x = -\frac{v_i^2}{2a}$ :  $v_i$  is squared, so doubling it multiplies  $\Delta x$  by  $2^2 = \boxed{4}$  ( $4 \times 60.0 \text{ m} = 240 \text{ m}$ ). Not 2.

### E3. Skyscraper: velocity as a function of time

**Why #1 (y-form):** it links  $v_y$  and  $t$  directly. Down is  $+$ .  $v_y = 0 + (+g)t \rightarrow \boxed{v_y(t) = +gt}$

**Why #2 (y-form) for the impact time:** you know  $\Delta y = +d$  and want  $t$ .  $+d = \frac{1}{2}gt_f^2 \rightarrow t_f = \sqrt{2d/g}$ , so it is valid for  $\boxed{0 \leq t \leq \sqrt{2d/g}}$ . Check:  $g\sqrt{2d/g} = \sqrt{2gd}$ , matching Question 4.

## Suggested partial credit (25 points per question, 100 total)

### 1. Acceleration-time graph

| Suggested partial credit                                                                                 | pts       |
|----------------------------------------------------------------------------------------------------------|-----------|
| Slope formula $a = \Delta v / \Delta t$ set up with values and units for each of the 4 segments (2 each) | 8         |
| Correct slope value with sign and units for each segment (1 each)                                        | 4         |
| Four horizontal segments at the correct accelerations (2 each)                                           | 8         |
| Changes in acceleration occur at the correct times; no slanted lines                                     | 2         |
| Axes labeled with quantity and units; readable scale                                                     | 3         |
| <b>Total</b>                                                                                             | <b>25</b> |

### 2. Braking car

| Suggested partial credit                                                         | pts       |
|----------------------------------------------------------------------------------|-----------|
| Starts from $v_f^2 = v_i^2 + 2a\Delta x$ (sheet #3) because time is not involved | 5         |
| Sets $v_f = 0$                                                                   | 3         |
| Acceleration negative, with a reason (opposite the motion / slowing down)        | 4         |
| Solves $\Delta x = -v_i^2 / (2a)$ before substituting                            | 6         |
| Substitution with units in every parenthesis                                     | 4         |
| Final answer, 3 significant figures, units, boxed                                | 3         |
| <b>Total</b>                                                                     | <b>25</b> |

### 3. Football range

| Suggested partial credit                                                           | pts       |
|------------------------------------------------------------------------------------|-----------|
| Resolves $v_0$ into $v_0 \cos \theta$ and $v_0 \sin \theta$                        | 4         |
| Time to peak from $v_y = 0$ with sheet #1: $t_{\text{peak}} = v_0 \sin \theta / g$ | 5         |
| Doubles $t_{\text{peak}}$ for the flight time, with the equal-height reason        | 4         |
| $R = v_{x0}t$ with the expression for $t$ substituted (not a rounded number)       | 5         |
| Substitution with units in every parenthesis                                       | 4         |
| Final answer, 3 significant figures, units, boxed                                  | 3         |
| <b>Total</b>                                                                       | <b>25</b> |

### 4. Skyscraper impact velocity

| Suggested partial credit                                                                    | pts       |
|---------------------------------------------------------------------------------------------|-----------|
| Writes an equation from the reference information first: $v_y^2 = v_{y0}^2 + 2a_y(y - y_0)$ | 4         |
| $v_{y0} = 0$ ; $a_y = +g$ and $\Delta y = +d$ , consistent with down positive               | 5         |
| Algebra to $v_y^2 = 2gd$                                                                    | 4         |
| Positive root with a reason (down is +); $v_y = +\sqrt{2gd}$ boxed                          | 3         |
| Answer uses only $d$ and $g$ ( $m$ correctly left out)                                      | 2         |
| B: indicates "consistent"                                                                   | 2         |
| B: justifies from the part A expression (speed $\propto \sqrt{g}$ , so $g/4$ gives half)    | 5         |
| <b>Total</b>                                                                                | <b>25</b> |